

MINECOFIN Rwanda fiscal projection and simulation model

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Overview

General properties

- Semi-structural "FPAS-type" gap model
- Five main blocks: demand, prices, monetary policy, exchange rate, fiscal policy
- Modeling demand components separately
- Four price indicators: core, food, energy, headline CPI
- Hybrid UIP with FX targeting
- Taylor rule
- Deficit targeting with automatic stabilizer effect

Linear gap models

- The focus is on the 2-5 year horizon
- Supply effects (through investment and capital) play a lesser role
- Real variables are decomposed into the sum of a gap and a trend component
- The model describes the behavior of the gap component, interpreted to capture short/medium term demand effects
- The trends represent the supply side, and are conceptually exogenous to the model

The new-Keynesian model

- The structure of gap models is motivated by the new-Keynesian model of a closed economy
- In this model
 - Inflation is related to the output gap, which is a proxy for the real marginal cost gap (the Phillips curve)
 - The output gap is related to the real interest rate gap, through the saving decisions of households (the IS curve)
 - An inflation targeting monetary policy stabilizes the economy (the policy rule)

The small open economy version

- Opening up the economy introduces foreign demand and imported inflation
- As well as the exchange rate and international financial markets
- Capital flows, consistency of real returns, and convergence to the rest of the world are major new issues
- Within the context of gap models, this leads to introduction of new trends: the real exchange rate trend, and the real interest rate trend
- Special features of the exchange rate arrangement of the particular country must be paid special attention

Fiscal policy

- The main purpose is to capture the effects of fiscal policy on output/inflation
- The main assumption is that fiscal policy has a deficit target in the long run
- The deficit in the long run is determined by systemic properties (such as the taxation rules, subsidization policy, efficiency of revenue collection, etc.)
- However, there can be temporary deviations, due to for example disaster relief, and other ad hoc spending decisions
- Obviously, there are also cyclical components (revenues), which vary without explicit action of the government
- This leads to a decomposition of the deficit, and then to the definition of the fiscal impulse, the main indicator of budgetary effects

Equations

General properties

- Both the real exchange rate and the real interest rate are defined in terms of foreign CPI and domestic core inflation
- Relative prices of domestic CPI components are with respect to overall CPI
- The inflation target is for overall CPI
- Foreign variables are described by the simplest possible equations (not listed here)
- All levels are in $100 * \log$ units, so a difference between two periods gives percentage change
- The frequency is quarterly
- All interest rates and quarterly growth rates are annualized

Equations

Demand

Prices

The exchange rate and monetary policy

Fiscal policy

Trend equations



Structure of the demand block

- Real GDP is decomposed into private consumption, private investment, government consumption, export, and import
- Government consumption includes government investment
- All five components are modeled as the sum of a gap and a trend
- Private consumption, private investment, and export gaps have their own behavioral equations
- Government consumption is exogenous
- The import gap is given as a combination of the other gaps, using weights based on import content
- Aggregate output gap and potential growth are weighted averages of the components

Private consumption gap

$$\begin{aligned} c\tilde{o}n s_t = & \alpha_1 c\tilde{o}n s_{t-1} + \alpha_2 c\tilde{o}n s_{t+1|t} - \alpha_3 r m c i_t^{cons} \\ & + \alpha_4 \tilde{y}_t + \alpha_5 f i s c_t + \varepsilon_t^{cons} \end{aligned} \quad (1)$$

- $c\tilde{o}n s_t$ is the private consumption gap
- $\tilde{y}_t$ is the aggregate output gap
- $r m c i_t^{cons}$ is the real monetary condition index for consumption

$$r m c i_t^{cons} = \alpha_6 \tilde{r}4_t + (1 - \alpha_6) \tilde{z}_t \quad (2)$$

- $\tilde{r}4_t$ is four-period-ahead forward looking average real interest rate gap

$$\tilde{r}4_t = (\tilde{r}_t + \tilde{r}_{t+1|t} + \tilde{r}_{t+2|t} + \tilde{r}_{t+3|t})/4 \quad (3)$$

- $\tilde{z}_t$ is the real exchange rate gap
- $f i s c_t$ is the fiscal impulse

Private investment gap

$$\begin{aligned} i\tilde{nv}_t = & \alpha_1 i\tilde{nv}_{t-1} + \alpha_2 i\tilde{nv}_{t+1|t} - \alpha_3 rmci_t^{inv} \\ & + \alpha_4 \tilde{y}_t + \alpha_5 fisc_t + \varepsilon_t^{inv} \end{aligned} \quad (4)$$

- Note that for simplicity of notation we use here the same parameter names as for the consumption gap; calibrated values are of course different
- $i\tilde{nv}_t$ is the private consumption gap
- $rmci_t^{inv}$ is the real monetary condition index for investment, given as

$$rmci_t^{inv} = \alpha_6 \tilde{r}_t + (1 - \alpha_6) \tilde{z}_t \quad (5)$$

Government consumption gap

$$gc\tilde{ons}_t = \alpha_1 gc\tilde{ons}_{t-1} + \varepsilon_t^{gcons} \quad (6)$$

- $gc\tilde{ons}_t$ is the government consumption+investment gap

Export gap

$$e\tilde{x}p_t = \alpha_1 e\tilde{x}p_{t-1} + \alpha_2 e\tilde{x}p_{t+1|t} - \alpha_3 rmci_t^{exp} + \alpha_5 \tilde{y}_t^* + \varepsilon_t^{exp} \quad (7)$$

- $e\tilde{x}p_t$ is the private consumption gap
- $rmci_t^{exp}$ is the real monetary condition index for investment, given as

$$rmci_t^{exp} = \alpha_6 \tilde{r}_t + (1 - \alpha_6) \tilde{z}_t \quad (8)$$

- $\tilde{y}_t^*$ is foreign output gap

Import gap

$$\begin{aligned} i\tilde{m}p_t = & w_{imp}^{cons} c\tilde{o}n s_t + w_{imp}^{inv} i\tilde{n}v_t + w_{imp}^{gcons} g\tilde{c}o n s_t \\ & + w_{imp}^{exp} e\tilde{x}p_t + \varepsilon_t^{imp} \end{aligned} \quad (9)$$

- $i\tilde{m}p_t$ is the import gap
- It is given as the weighted average of the other four demand components
- The weights are determined using the import content of respective component
- The shock captures, among other things, the variations in the import contents

Aggregate output gap

$$\begin{aligned}\tilde{y}_t = & w_y^{cons} c\tilde{o}n s_t + w_y^{inv} i\tilde{n}v_t + w_y^{gcons} g c\tilde{o}n s_t \\ & + w_y^{exp} e\tilde{x}p_t - w_y^{imp} i\tilde{m}p_t + \varepsilon_t^y\end{aligned}\quad (10)$$

- $\tilde{y}_t$ is the aggregate output gap
- It is given as the linear combination of the demand components
- The weights are determined using the nominal shares of the components
- The shock captures, among other things, the variations in the shares

Equations

Demand

Prices

The exchange rate and monetary policy

Fiscal policy

Trend equations



Structure of the price block

- Three price variables: core, food, and energy prices
- All three have their separate homogeneous Phillips curves
- Imported inflation appears in all three, both with immediate and with gradual pass-through
- Relative price effects ensure that all three inflation rates are stabilized by the central bank
- But not all three inflations rates are equal in the steady state
 - Food prices have been growing faster the CPI over the history
- Several trend adjustments are employed to remove real trend effects from the Phillips curves

Core Phillips curve

$$\pi_t^{\text{core}} = \beta_1 \pi_{t-1}^{\text{core}} + (1 - \beta_1 - \beta_3) \pi_{t+1|t}^{\text{core}} + \beta_3 (\pi_t^* + \Delta s_t - \Delta \bar{z}_t) + \beta_2 rmc_t + \varepsilon_t^{\text{core}} \quad (11)$$

- π_t^{core} is core inflation
- π_t^* is foreign CPI inflation
- Δs_t is nominal exchange rate change
- $\Delta \bar{z}_t$ is trend real exchange rate change
- rmc_t is the real marginal cost, given as

$$rmc_t = \beta_4 \tilde{y}_t + (1 - \beta_4) \tilde{z}_t \quad (12)$$

Food Phillips curve

$$\begin{aligned}\pi_t^{\text{food}} = & \beta_{f1} \pi_{t-1}^{\text{food}} + (1 - \beta_{f1} - \beta_{f3}) \pi_{t+1|t}^{\text{food}} + \\ & + \beta_{f3} \left(\pi_t^{\text{food*}} - \Delta \overline{rp}_t^{\text{food*}} + \Delta s_t - \Delta \bar{z}_t - \Delta \overline{rp}_t^{\text{core}} + \Delta \overline{rp}_t^{\text{food}} \right) + \\ & + \beta_{f2} \left(\tilde{rp}_t^{\text{food*}} + \tilde{z}_t + \tilde{rp}_t^{\text{core}} - \tilde{rp}_t^{\text{food}} \right) + \varepsilon_t^{\text{food}}\end{aligned}\quad (13)$$

- π_t^{food} is food inflation
- $\pi_t^{\text{food*}}$ is world food price inflation (in USD)
- $\Delta \overline{rp}_t^{\text{food*}}$ is trend world relative food price change
- $\Delta \overline{rp}_t^{\text{core}}$ and $\Delta \overline{rp}_t^{\text{food}}$ are trend domestic relative (to CPI) core and food price changes
- $\tilde{rp}_t^{\text{food*}}$ is world relative food price gap
- $\tilde{rp}_t^{\text{core}}$ and $\tilde{rp}_t^{\text{food}}$ are domestic relative core and food price gaps

Energy Phillips curve

$$\begin{aligned} \pi_t^{\text{ener}} = & \beta_{f1} \pi_{t-1}^{\text{ener}} + (1 - \beta_{f1} - \beta_{f3}) \pi_{t+1|t}^{\text{ener}} + \\ & + \beta_{f3} \left(\pi_t^{\text{ener}*} - \Delta \overline{rp}_t^{\text{ener}*} + \Delta s_t - \Delta \bar{z}_t - \Delta \overline{rp}_t^{\text{core}} + \Delta \overline{rp}_t^{\text{ener}} \right) + \\ & + \beta_{f2} \left(\tilde{rp}_t^{\text{ener}*} + \tilde{z}_t + \tilde{rp}_t^{\text{core}} - \tilde{rp}_t^{\text{ener}} \right) + \varepsilon_t^{\text{ener}} \end{aligned} \quad (14)$$

- π_t^{ener} is energy inflation
- $\pi_t^{\text{ener}*}$ is world oil price inflation (in USD)
- $\Delta \overline{rp}_t^{\text{ener}*}$ is trend world relative oil price change
- $\Delta \overline{rp}_t^{\text{core}}$ and $\Delta \overline{rp}_t^{\text{ener}}$ are trend domestic relative (to CPI) core and energy price changes
- $\tilde{rp}_t^{\text{ener}*}$ is world relative food price gap
- $\tilde{rp}_t^{\text{core}}$ and $\tilde{rp}_t^{\text{ener}}$ are domestic relative core and energy price gaps

Headline CPI

$$\pi_t = w_{\pi}^{\text{core}} \pi_t^{\text{core}} + w_{\pi}^{\text{food}} \pi_t^{\text{food}} + w_{\pi}^{\text{ener}} \pi_t^{\text{ener}} + \varepsilon_t^{\pi} \quad (15)$$

- π_t is headline CPI inflation
- It is given as the weighted average of the demand components
- The weights are the official CPI weights
- The shock captures, among other things, the variations in the weights

Equations

Demand

Prices

The exchange rate and monetary policy

Fiscal policy

Trend equations



The monetary and FX regime

- Monetary policy is a combination of inflation and exchange rate targeting
- The exchange rate target depreciation rate is consistent with the PPP in the long run
- But it is flexible to incorporate cyclical deviation
- In particular, both the interest rate and the exchange rate target react to inflation
- The actual exchange rate is a weighted average of the UIP consistent and the target rate
- And the UIP itself incorporates both forward and backward looking expectations

Hybrid UIP

$$s_t = \eta_1 (s_t^{\text{exp}} - (i_t - i_t^* - \text{prem}_t) / 4) + (1 - \eta_1) (s_{t-1} + \Delta s_t^{\text{tar}} / 4) + \varepsilon_t^s \quad (16)$$

- s_t is the USDRWF exchange rate (depreciation is up)
- i_t^* is the Fed Funds Rate
- Δs_t^{tar} is the target nominal depreciation
- s_t^{exp} is the expected exchange rate, given as

$$s_t^{\text{exp}} = \eta_2 (s_{t-1} + 2 (\Delta \bar{z}_t + \bar{\pi} 4_t + \Delta \bar{r} \bar{p}_t^{\text{core}} - \pi^*) / 4) + (1 - \eta_2) s_{t+1|t} \quad (17)$$

The target exchange rate

$$\begin{aligned}
 \Delta s_t^{\text{tar}} = & \eta_4 \Delta s_{t-1}^{\text{tar}} + (1 - \eta_4) \left(\Delta \bar{z}_t + \overline{\pi 4}_t + \Delta \overline{rp}^{\text{core}} - \pi^* \right. \\
 & \left. + \eta_5 \left(\pi 4_{t+4|t} - \overline{\pi 4}_t \right) - \eta_6 \tilde{z}_t \right)
 \end{aligned} \tag{18}$$

- The target depreciation rate is consistent with PPP in the long run
- It also incorporates inflation targeting and exchange rate stabilization motives on the short run

Monetary policy

$$i_t = \gamma_1 i_{t-1} + (1 - \gamma_1) \left(\bar{i}_t + \gamma_2 \left(\pi_{4_{t+4}|t} - \overline{\pi}_{4_t} \right) + \gamma_3 \tilde{y}_t + \gamma_4 (\Delta s_t - \Delta s_t^{\text{tar}}) \right) + \varepsilon_t^i \quad (19)$$

- Inflation forecast based (IFB) policy rule
- With possible FX targeting and direct demand control motives
- The neutral rate $\bar{i}_t$ is the sum of the inflation target and the trend real interest rate

$$\bar{i}_t = \bar{r}_t + \overline{\pi}_{4_t} + \Delta \overline{r p}_t^{\text{core}} \quad (20)$$

- The real interest rate trend is based the UIP

$$\bar{r}_t = \bar{r}_t^* + \text{prem}_t - \Delta \bar{z}_{t+1|t} \quad (21)$$

- Using a "UIP-like" relationship between trend variables is actually a behavioral assumption

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Prices

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Deficit decomposition

$$def_t = def_t^{\text{struct}} + def_t^{\text{cyc}} + def_t^{\text{discr}} \quad (22)$$

- def_t is budget deficit, in % of GDP, without grant revenues
- def_t^{struct} is the structural deficit
- def_t^{cyc} is the cyclical component, a.k.a the automatic stabilizer
- def_t^{discr} is the discretionary (or idiosyncratic) component

The cyclical component

$$def_t^{cyc} = \delta_1 \tilde{y}_t \quad (23)$$

- def_t^{cyc} is interpreted as capturing the *revenue effects* of the economic cycle
- In the long the cyclical component is zero, as the economy follows its long run equilibrium path

The structural component

$$def_t^{struct} = \delta_4 def_{t-1}^{struct} + (1 - \delta_4) \overline{def}^{struct} + \delta_6 (\Delta \overline{gcons}_t - \Delta \bar{y}_t) + \varepsilon_t^{def^{struct}} \quad (24)$$

- def_t^{struct} is deficit corresponding to the long term equilibrium path of the economy
- $\Delta \overline{gcons}_t$ is the trend growth of government consumption
- $\Delta \bar{y}_t$ is potential output growth

The discretionary component

$$def_t^{\text{discr}} = \delta_2 def_{t-1}^{\text{discr}} + \delta_5 gc\tilde{ons}_t + \varepsilon_t^{def\text{discr}} \quad (25)$$

Fiscal impulse

$$fisc_t = \delta_3 def_t^{\text{discr}} + (1 - \delta_3) \Delta def_t^{\text{struct}} \quad (26)$$

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Demand

Prices

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Trend equations



General form of trend equations

$$\Delta \bar{x}_t = \rho_{\bar{x}} \Delta \bar{x}_{t-1} + (1 - \rho_{\bar{x}}) \Delta \bar{x} + \varepsilon_t^{\bar{x}} \quad (27)$$

- There is a steady state growth rate of the trend
- After any deviation from this growth rate, there's a slow adjustment
- We have this form for the five demand components, the real exchange rate, and the domestic relative prices of food and energy

Aggregate trend growth

$$\Delta \bar{y}_t = w_y^{cons} \Delta \bar{cons}_t + w_y^{inv} \Delta \bar{inv}_t + w_y^{gcons} \Delta \bar{gcons}_t + w_y^{exp} \Delta \bar{exp}_t - w_y^{imp} \Delta \bar{imp}_t \quad (28)$$

- Potential growth is a linear combination of the trend growth rates of the components
- Together with the analogous equation for the output gap, this implies

$$\Delta y_t = w_y^{cons} \Delta cons_t + w_y^{inv} \Delta inv_t + w_y^{gcons} \Delta gcons_t + w_y^{exp} \Delta exp_t - w_y^{imp} \Delta imp_t + \Delta \varepsilon_t^y$$

Relative core price trend

- The CPI aggregation equation implies that

$$0 = w_{\pi}^{\text{core}} \Delta r p_t^{\text{core}} + w_{\pi}^{\text{food}} \Delta r p_t^{\text{food}} + w_{\pi}^{\text{ener}} \Delta r p_t^{\text{ener}}$$

- We further require that

$$0 = w_{\pi}^{\text{core}} \tilde{r} p_t^{\text{core}} + w_{\pi}^{\text{food}} \tilde{r} p_t^{\text{food}} + w_{\pi}^{\text{ener}} \tilde{r} p_t^{\text{ener}} \quad (29)$$

- These two equations imply that the trend of relative core price is given as a residual, from the following equation

$$0 = w_{\pi}^{\text{core}} \Delta \overline{r} p_t^{\text{core}} + w_{\pi}^{\text{food}} \Delta \overline{r} p_t^{\text{food}} + w_{\pi}^{\text{ener}} \Delta \overline{r} p_t^{\text{ener}}$$

Calibration

Calibration principles

- The model parameters have been calibrated such that
 - the model has economically sensible simulation properties
 - the estimated unobserved variables and shocks are in line with expert knowledge about the history of the Rwandan economy
- Besides, we have used the parameterizations of other models as a reference
 - BNR
 - Ghana (GH), South Africa (SARB)
 - Hungary (HU) and Armenia (AM) (these models have demand decomposition)

Private consumption

Name	Description	Value	Justification
α_1^{cons}	Lag	0.65	Close to HU and AM models
α_2^{cons}	Expectation	0.00	Same as GH model
α_3^{cons}	RMCI	0.10	Between to HU and AM models
α_4^{cons}	Income	0.30	Close to AM model
α_5^{cons}	Fiscal impulse	0.50	Historical data
α_6^{cons}	Domestic component in RMCI	1.00	Theory

Private investment

Name	Description	Value	Justification
α_1^{cons}	Lag	0.65	Close to HU and AM models
α_2^{cons}	Expectation	0.00	Same as GH model
α_3^{cons}	RMCI	0.10	Between to HU and AM models
α_4^{cons}	Income	0.30	Close to AM model
α_5^{cons}	Fiscal impulse	0.30	Historical data, likely less than for consumption
α_6^{cons}	Domestic component in RMCI	1.00	Theory

Government consumption

Name	Description	Value	Justification
α_1^{gcons}	Lag	0.65	Same as for the other components

Export

Name	Description	Value	Justification
α_1^{exp}	Lag	0.65	Close to HU and AM models
α_2^{exp}	Expectation	0.00	Same as GH model
α_3^{exp}	RMCI	0.10	A bit larger to HU and AM models
α_5^{exp}	Foreign demand	0.50	Between HU and AM models

Import

Name	Description	Value	Justification
w_{imp}^{cons}	Private consumption gap	0.23	Expert knowledge
w_{imp}^{inv}	Private investment gap	0.31	Expert knowledge
w_{imp}^{gcons}	Government consumption gap	0.31	Expert knowledge
w_{imp}^{exp}	Export gap	0.15	Expert knowledge

GDP

Name	Description	Value	Justification
w_y^{cons}	Private consumption gap	0.78	Nominal share in recent years
w_y^{inv}	Private investment gap	0.13	Nominal share in recent years (residual)
w_y^{gcons}	Government consumption gap	0.23	Nominal share in recent years
w_y^{exp}	Export gap	0.21	Nominal share in recent years
w_y^{imp}	Import gap	0.35	Nominal share in recent years

Core Phillips curve

Name	Description	Value	Justification
β_1	Lag	0.35	Smaller than in the references, but gives better IRF and historical forecasts
β_2	RMC	0.20	Based on BNR
β_3	Direct foreign	0.05	Close to SARB
β_4	Domestic in RMC	0.80	Close to references

Food Phillips curve

Name	Description	Value	Justification
β_{f1}	Lag	0.50	Close to references
β_{f2}	RMC	0.10	Close to references
β_{f3}	Direct foreign	0.05	Close to SARB
β_{f4}	Domestic in RMC	0.00	Expert judgment

Energy Phillips curve

Name	Description	Value	Justification
β_{e1}	Lag	0.50	Close to references
β_{e2}	RMC	0.01	Historical decomposition, smaller the in the references
β_{e3}	Direct foreign	0.01	Historical decomposition, smaller the in the references
β_{e4}	Domestic in RMC	0.00	Expert judgment

Exchange rate

Name	Description	Value	Justification
η_1	Interest differential in UIP	0.50	Taken from BNR
η_2	Forward in expectations	0.20	Taken from BNR
η_4	Persistence of FX target	0.00	Taken from BNR
η_5	Inflation in FX target	0.30	Taken from BNR
η_5	RER gap in FX target	0.85	Taken from BNR

Monetary policy rule

Name	Description	Value	Justification
γ_1	Lag	0.50	Smaller than references
γ_2	Inflation reaction	0.50	Lower than references, except BNR
γ_3	Output gap	0.50	Similar to references (but BNR is smaller)
γ_4	FX target	0.00	As in the references



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